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Real-Time Air Quality Monitor and Predictor

(User Manual)

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Spring 2026

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Setup/Installation Guide

Pre-installation Tasks

The user checks that the computer meets the software requirements before installation. A stable internet connection, Node.js, and a supported web browser are available on the system. The user ensures enough storage space is free and confirms that all required dependencies are ready for installation. Proper permissions are also available to install and run the application successfully.

Installation Procedure

The user downloads the application files and extracts them to a suitable location on the computer. The application folder is opened in Visual Studio Code, and the required packages are installed using the npm install command. After successful package installation, the user starts the application through the terminal. The browser opens automatically and displays the main interface for access to system features.

Post-installation Tasks

The user signs in with valid credentials and verifies that all modules are functioning correctly. Air quality information, prediction services, notifications, and dashboard features are checked to ensure proper operation. Profile settings may be updated according to user preferences. A brief test of key functions confirms that the application is ready for regular use.

Uninstall or Rollback

The user removes the application through the operating system's uninstall option or by deleting the application directory from the computer. Any remaining temporary files, cached data, and unused dependencies may also be removed to free storage space. Before removal, the user backs up important data if it needs to be retained. For rollback, the user restores a previously saved version of the application from a backup. The older version is installed, and the associated configuration files and database backups are recovered if required. After restoration, the user verifies that all features operate correctly and that stored data remains accessible.

User Guide

Launching the Application

The user begins by signing in to the application using a registered account. After successful authentication, the dashboard displays air quality information, pollution levels, weather details, and prediction results. Navigation options allow access to different sections, including historical data, alerts, reports, profile settings, and feedback features. The user can view air quality measurements through charts, graphs, and maps. Notifications are displayed when pollutant levels exceed predefined thresholds. Historical records help users analyze trends and compare air quality data over different periods. Export options enable reports to be saved for further analysis or documentation purposes. The profile section allows users to update personal information and manage notification preferences. Feedback and complaint features enable users to submit suggestions or report issues directly through the application. The interface is designed to provide easy access to information and support informed environmental decision-making.

Dashboard Overview

Figure 1 shows the Dashboard Overview page of the Quality Monitor system. The dashboard provides a summary of system status, performance metrics, alerts, and quick access to key monitoring and management functions

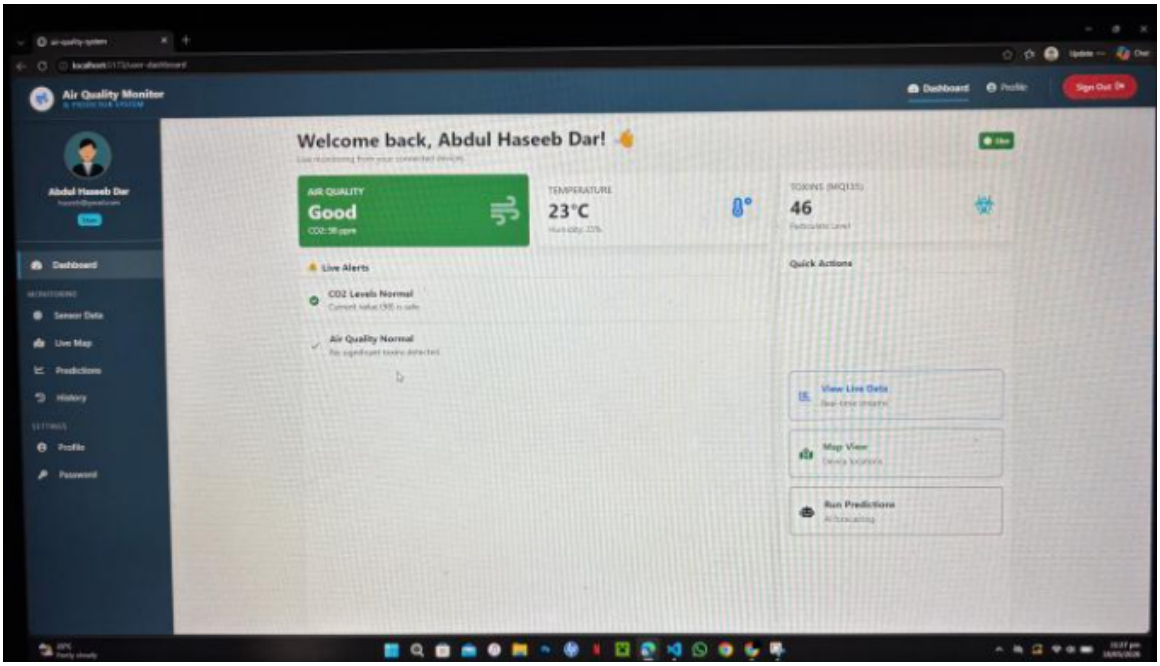


Figure 1: Desktop View of Dashboard Overview Screen

Login Screen

[Figure 2] presents the Login Screen used for user authentication. Users enter valid credentials to gain access to the system. Successful authentication redirects users to the dashboard, while invalid credentials generate an appropriate error message Figure 2 undefined

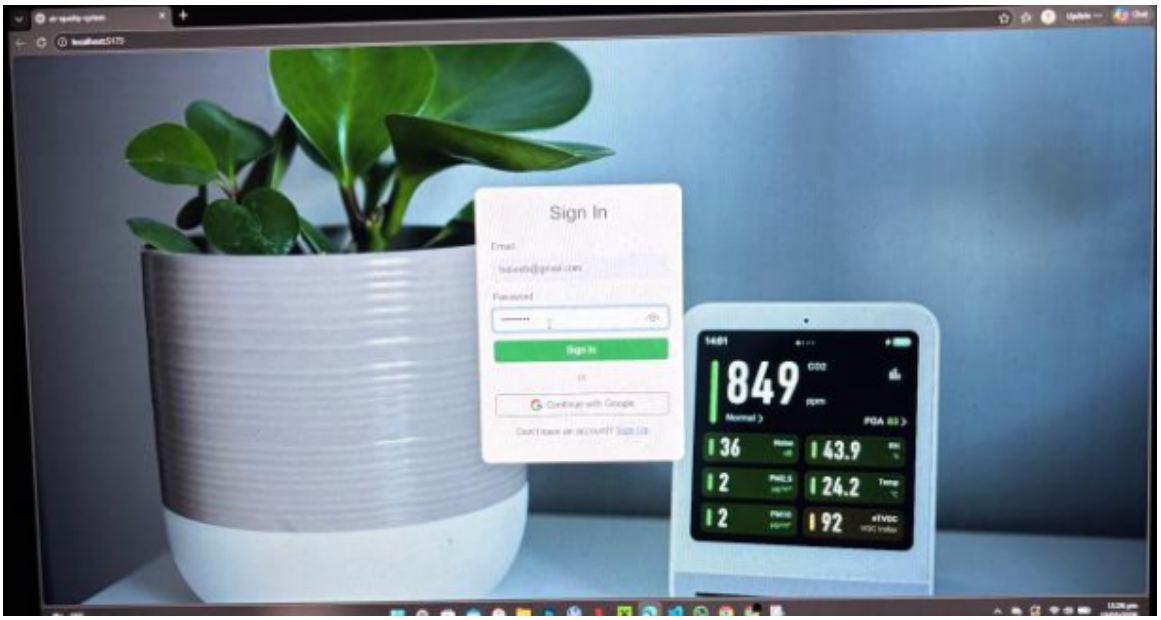


Figure 2: User Login interface requiring valid credentails before system access

Live Maps

Figure 3 displays the Live Map interface. Sensor locations are plotted on a geographic map, allowing users to view air quality readings across different regions. Cluster markers improve visualization by grouping nearby sensors and simplifying data interpretation

Sensor Analyrics

Figure 4 illustrates the Sensor Analytics interface. The page provides detailed insights from collected sensor measurements through charts, graphs, and statistical indicators. Users can evaluate trends, compare readings, and identify variations in air quality across monitored locations.

Prediction Screen

Figure 5 Presents the Prediction interface. Users enter environmental parameters such as temperature, humidity, and carbon dioxide concentration. The prediction model processes the input values and generates an estimated air quality classification based on learned patterns from historical sensor data

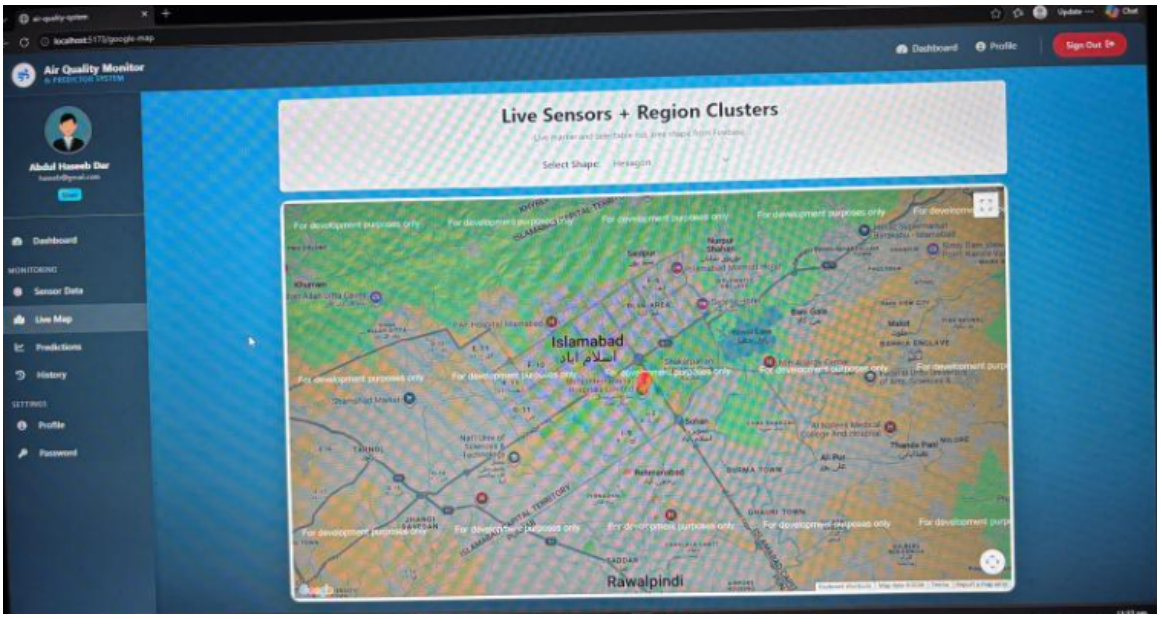


Figure 3: Live Map showing sensor locations and regional clusters for air quality monitoring

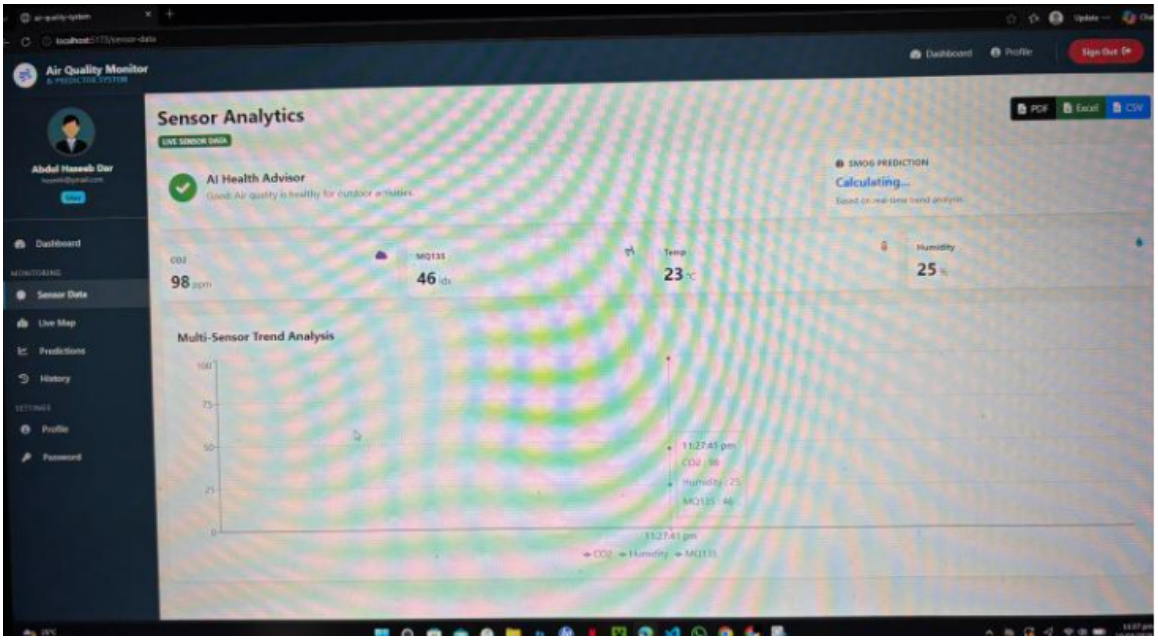


Figure 4: Analytics view of sensors data using charts and performance indicators

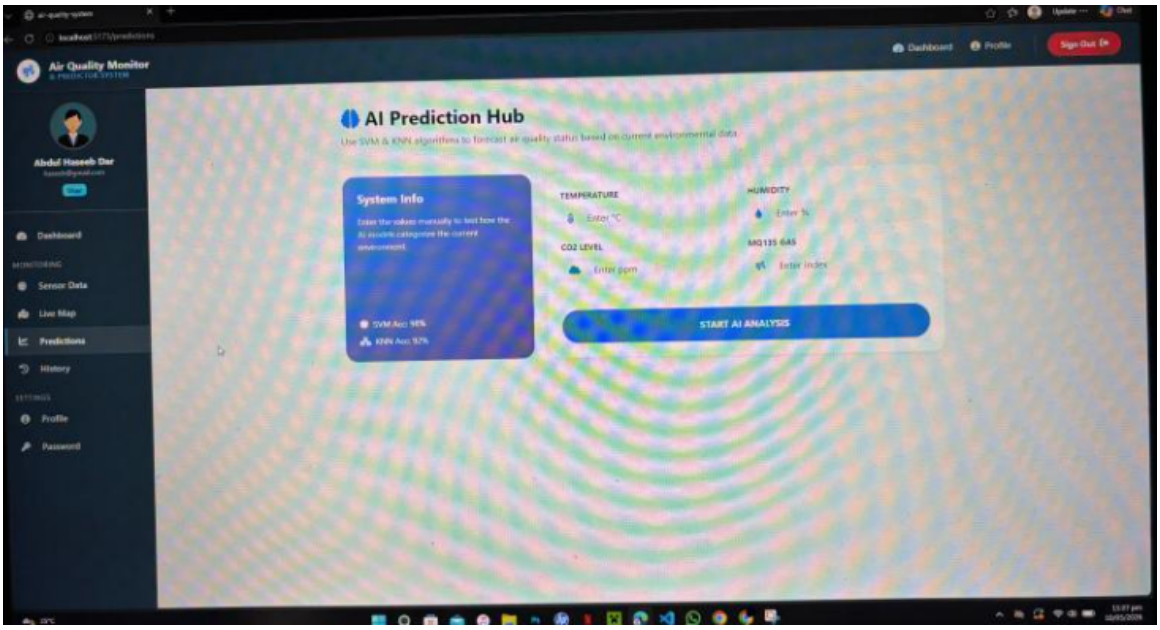


Figure 5: Prediction Interface using AI for estimating air quality using environmental parameters